

2: Variables, Expressions, and Statements

2.1: Values and Types

A value is one of the basic things a program works with, like a letter or a number. The values we have seen so far are 1, 2, and "Hello, World!"

2.3: Variable names and Keywords

Programmers generally choose names for their variables that are meaningful and document what the variable is used for. Variable names can be arbitrarily long. They can contain both letters and numbers, but they cannot start with a number. It is legal to use uppercase letters, but it is a good idea to begin variable names with a lowercase letter.

2.5: Operators and Operands

Operators are special symbols that represent computations like addition and multiplication. The values the operator is applied to are called operands.

2.7: Order of Operations

When more than one operator appears in an expression, the order of evaluation depends on the rules of precedence. For mathematical operators, Python follows mathematical convention.

2.9: String Operations

The + operator works with strings, but it is not addition in the mathematical sense. Instead it performs **concatenation**, which means joining the strings by linking them end to end.

2.G: Glossary

2.11: Comments

As programs get bigger and more complicated, they get more difficult to read. Formal languages are dense, and it is often difficult to look at a piece of code and

2.2: Variables

One of the most powerful features of a programming language is the ability to manipulate variables. **A variable is a name that refers to a value. An assignment statement creates new variables and gives them values.**

2.4: Statements

A statement is a unit of code that the Python interpreter can execute. We have seen two kinds of statements: print being an expression statement and assignment.

2.6: Expressions

An expression is a combination of values, variables, and operators. A value all by itself is considered an expression, and so is a variable, so the following are all legal expressions (assuming that the variable x has been assigned a value):

2.8: Modulus Operator

The modulus operator works on integers and yields the remainder when the first operand is divided by the second. In Python, the modulus operator is a percent sign (%). The syntax is the same as for other operators.

2.E: Variables, Expressions, and Statements (Exercises)

2.10: Asking the user for input

Sometimes we would like to take the value for a variable from the user via their keyboard. Python **provides a built-in function called input that gets input from the keyboard.** When this function is called, the program stops and waits for the user to type something. When the user presses Return or Enter, the program resumes and input returns what the user typed as a string.

2.12: Choosing Mnemonic Variable Names

As long as you follow the simple rules of variable naming, and avoid reserved words, you have a lot of choice when you name your variables. In the beginning,

figure out what it is doing, or why. For this reason, it is a good idea to add notes to your programs to explain in natural language what the program is doing. These notes are called comments, and in Python they start with the # symbol.

this choice can be confusing both when you read a program and when you write your own programs. For example, the following three programs are identical in terms of what they accomplish, but very different when you read them and try to understand them.

2.13: Debugging

At this point, the syntax error you are most likely to make is an illegal variable name, like class and yield, which are keywords, or odd~job and US\$, which contain illegal characters.

This page titled [2: Variables, Expressions, and Statements](#) was last modified on Tue, 01 Dec 2020 03:48:10 GMT and is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by [Chuck Severance](#) via [source content](#) that was edited to the style and standards of the LibreTexts platform.